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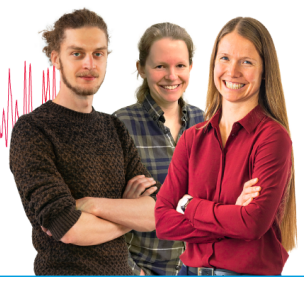
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The $D\,^1\Sigma^+$ state of ^{7}LiH: Comparison of observations with vibronic theory

Chia-Ying Wu,[1] Wei-Tzou Luh,[1,a)] F. Xavier Gadea,[2] and William C. Stwalley[3]
[1]*Department of Chemistry, National Chung Hsing University, 250 Kuo-Kuang Road, Taichung 402, Taiwan*
[2]*Laboratoire de Physique Quantique, IRSAMC, Universite Paul Sabatier, 118 route de Narbonne, 31062 Toulouse Cedex 4, France*
[3]*Department of Physics, University of Connecticut, 2152 Hillside Road, Storrs, Connecticut 06269-3046, USA*

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The excited $D\,^1\Sigma^+$ electronic state of ^{7}LiH has been observed up to near its dissociation limit by a pulsed optical-optical double resonance fluorescence depletion spectroscopic technique. An extensive vibronic calculation has been performed with a diabatic approach with purely potential couplings involving a set of eight diabatic states of $^1\Sigma^+$ symmetry, corresponding to seven neutral states and one ionic state. Twenty-six new vibrational levels have been observed. Both the derived vibrational energy spacings and the vibronic ones are similarly irregular. The observed spectral linewidths and vibronic resonance widths are found to vary similarly with increasing energy. Observed asymmetric spectral lineshapes may be attributed to the strong radial couplings between the discrete levels of the $D\,^1\Sigma^+$ electronic state and the continuum states of the $C\,^1\Sigma^+$ electronic state. The mutual agreement between the spectral results and the vibronic results demonstrates that the $D\,^1\Sigma^+$ electronic state of ^{7}LiH is better characterized by the vibronic approach. © *2008 American Institute of Physics*. [DOI: 10.1063/1.2827132]

I. INTRODUCTION

Lithium hydride, the simplest neutral heteronuclear molecule, has been the object of intensive spectroscopic[1–21] and theoretical[22–31] studies since the 1930s. The lowest five singlet electronic states, $X\,^1\Sigma^+$, $A\,^1\Sigma^+$, $B\,^1\Pi$, $C\,^1\Sigma^+$, and $D\,^1\Sigma^+$, have been identified and characterized by conventional spectroscopy[1–7] and by laser spectroscopy.[8–21] For brevity, in Fig. 1, we portray only the three relevant potentials, an effective hybrid potential of the $C\,^1\Sigma^+$ state,[18] an adiabatic *ab initio* potential of the $D\,^1\Sigma^+$ state,[19,30] and a Li^+H^- ionic curve which includes only the R^{-1} term.

In an effective core potential calculation,[27] Boutalib and Gadea reported that the potential energy curves for highly excited electronic states of LiH would become complex due to the avoided crossings between the valence and Rydberg atomic configurations on the one hand and between the ionic and the neutral configurations on the other hand. The first intriguing double-well C electronic state, among three different isotopomers, ^{6}LiH, ^{7}LiH, and ^{7}LiD, was investigated by Luh *et al.*[17,18,20] using a pulsed optical-optical double resonance (OODR) fluorescence depletion spectroscopic technique. In a recent sub-Doppler cw-OODR experiment, Bouloufa *et al.*[21] performed lifetime measurements in the C state. Nonradiative lifetimes among 11 measured vibrational levels were found to be systematically shorter than the theoretical ones resulting from a vibronic calculation,[29] but they do follow a similar variation with respect to the vibrational quantum number.

In a vibronic calculation for the double-well D state,[30] Gemperle and Gadea predicted that, (i) for the first time, the adiabatic Born-Oppenheimer approximation had completely broken down for a diatomic molecule, (ii) the vibronic shift was quite large, and (iii) the calculated resonance widths [half width at half maximum (HWHM)] varied over the range of $4\times10^{-5}-38$ cm^{-1}. Huang *et al.*[19] presented the first OODR results on the bottom part of the potential of the D state of ^{7}LiH. An overall measure of three different aspects, the energetics, the spectral line shapes, and the transition probabilities, strongly favors the argument that the vibronic approach, which reflects a complete breakdown of the Born-Oppenheimer approximation, is a better approach for characterizing the excited D electronic state.

In this work, we have observed the upper part of the D state after the calibration technique for the blue spectral range with a tellurium-130 cell[32] became available in our laboratory. An extensive vibronic calculation has been performed with a diabatic approach with purely potential couplings involving a set of eight diabatic states of $^1\Sigma^+$ symmetry, corresponding to seven neutral asymptotes $[Li(2s,2p,3s,3p,3d,4s,4p)H(1s)]$ and one ionic asymptote (Li^+H^-).[30] The spectral results are compared with the adiabatic and vibronic results among all likely vibrational levels of the D state.

II. EXPERIMENT

The experimental setup is similar to that reported previously.[19] The pump laser system was a frequency-doubled Nd:YAG (YAG denotes-yttrium aluminum garnet) laser (Lumonics YM 600) pumped dye laser (Lumonics HD500). The Q-switched Nd:YAG laser operated at 10 Hz repetition rate with a 7 ns pulse length and 25 mJ pulse energy. The dye laser beam was frequency-doubled by a frequency-doubling system (Inrad autotracker II) to generate

a)Author to whom the correspondence should be addressed. Electronic mail: wtluh@dragon.nchu.edu.tw.

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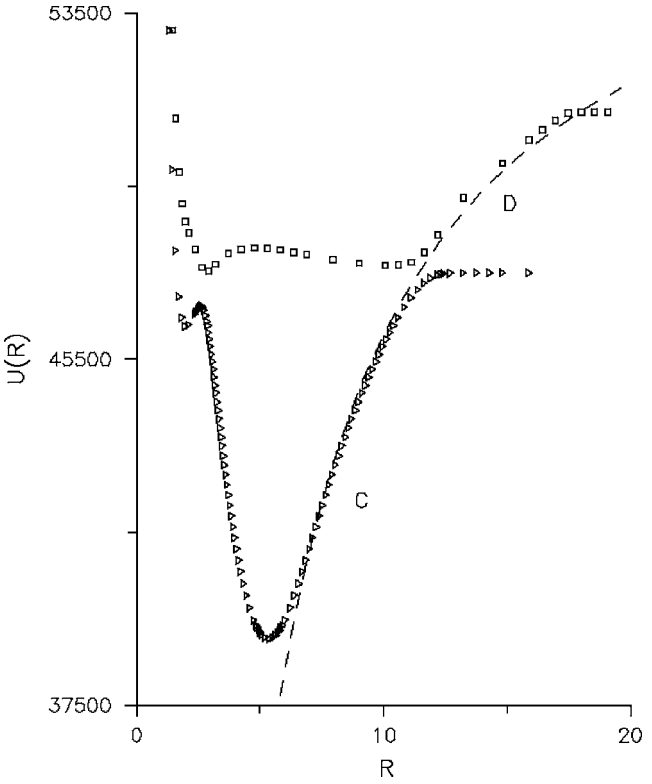


FIG. 1. Potential energy curves of ^{7}LiH: an effective hybrid $C\ ^1\Sigma^+$ potential (open triangles, from Ref. 18), and an *ab initio* $D\ ^1\Sigma^+$ potential (open squares, from Ref. 19). The dashed line represents the Li^+H^- ionic curve. U(R) in cm^{-1} and *R* in angstroms.

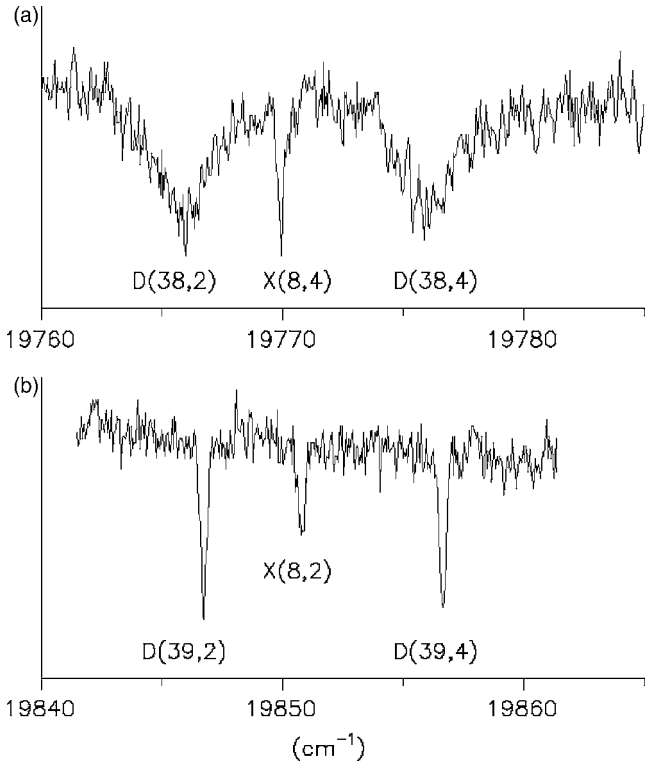


FIG. 2. OODR spectra of ^{7}LiH. The rovibrational energy level $A\ ^1\Sigma^+(10,3)$ was the intermediate state.

an ultraviolet PUMP laser beam for exciting the LiH molecule from the $X\ ^1\Sigma^+$ state to the $A\ ^1\Sigma^+$ state. A methanol solution of a dye (LDS698) was used as the active medium for the spectral range of the A(7-12)-X(0) bands. The probe laser was a frequency doubled/tripled Nd:YAG laser (Quantel BW) pumped dye laser (Lumonics HD500). The *Q*-switched Nd:YAG laser operated at 10 Hz repetition rate with a 4 ns pulse length and 20 mJ pulse energy. Several laser dyes (C460, C480, C503, and C540A) dissolved in methanol were used for the *D-A* OODR spectral range, 17 390–23 000 cm^{-1}, calibrated with an I_2 or Te_2 reference spectrum.[32] The overall uncertainty of the frequency measurements is ~0.1 cm^{-1}. The pump laser beam and the probe laser beam were directed collinearly along the axis of the longer arms of a five-arm crossed heat-pipe oven. The time delay between the probe laser pulse and the pump laser pulse was regulated with two digital delay/pulse generators (Stanford DG535). The heat-pipe oven, containing 10 g of naturally abundant lithium metal (STREM, 3N) in its center, was typically operated under an argon gas of one Torr and at the temperature of 953 K. The laser induced *A-X* fluorescence was detected by a photomultiplier detector (Hamamatsu R928) after a spectrometer, having a spectral bandpass [full width at half maximum (FWHM)] of 9 nm. The detector signal was fed into a boxcar-averager-gated-integrator system (Stanford SRS250), and the averaged output was then recorded by a personal computer. The OODR fluorescence depletion spectrum was obtained by monitoring the variation of the laser induced *A-X* fluorescence intensity as the probe laser wavelength was scanned, while the pump laser was fixed at a particular LiH *A-X* rovibrational transition.

III. RESULTS AND DISCUSSION

A. OODR spectra

For the pump transitions among the $A\ ^1\Sigma^+ - X\ ^1\Sigma^+$ electronic transitions, we selected those vibrational bands having known favorable radiative transition probabilities.[33] The likely spectral ranges and radiative transition probabilities for the probe transitions were calculated[19] and used as a reference.

The rovibrational energy levels of $A\ ^1\Sigma^+(v'=6-12, J'=0-10)$ were selected as intermediate states. The electric polarization of the probe laser beam was oriented parallel to that of the UV pump laser beam. With such an arrangement, we occasionally obtained a spectrum covering both the OODR-type transitions up to the excited $D\ ^1\Sigma^+$ state and the stimulated emission pumping (SEP)-type transitions[34] down to the ground $X\ ^1\Sigma^+$ electronic state. Two such OODR spectra out of the $A\ ^1\Sigma^+(v'=10, J'=3)$ intermediate state are presented in Fig. 2 and will be further discussed in Sec. III C 2. Note that the linewidth(~0.2 cm^{-1}, FWHM) of the observed SEP-type transition would reflect the narrowest linewidth obtainable in our OODR scheme.

B. Rotational and vibrational constants

We observed 292 new OODR transitions, corresponding to 197 rovibrational levels among 26 vibrational levels in the $D\ ^1\Sigma^+$ state. The term energy for each observed rovibrational level is obtained by adding together the probe frequency and

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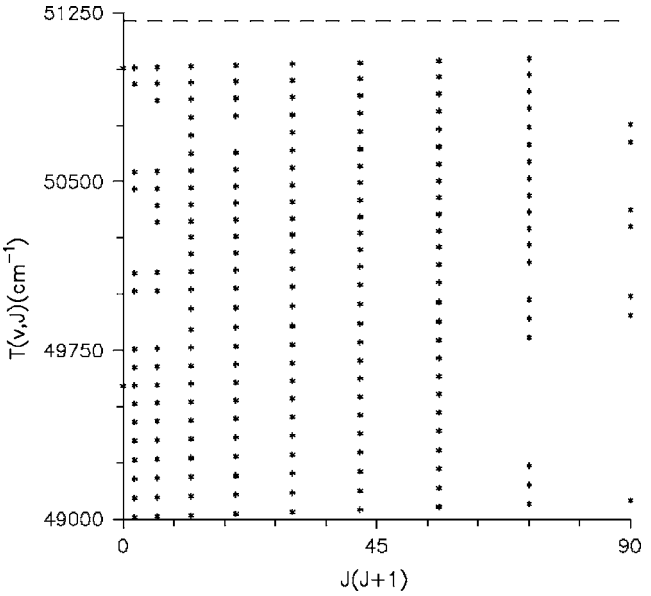


FIG. 3. Plot of 192 observed term energies $T(v,J)$ (in cm^{-1}) of the ($J=0-9$) rovibrational levels among the vibrational levels $v=26-51$ of the ^{7}LiH $D\,^1\Sigma^+$ state, with respect to the $J(J+1)$ product. The upper dashed line represents the energy of the dissociation limit, H($1s$)+Li($3p$), of the $D\,^1\Sigma^+$ state.

the term energy of the intermediate rovibrational level, which was calculated with known Dunham-type coefficients.[35] It is noted that, for the $D(47,J')-A(8,2-8)$ and $D(48,J')-A(8,2-7)$ OODR transitions, the P and R transitions were found not well separated and some even overlapped as a single broad peak, particularly out of the low J levels. The term energies for the five rovibational levels, $D(47,J'=1-2,4)$ and $D(48,J'=1-2)$, were not estimated. The 192 term energies for the rovibrational levels, $26\leqslant v\leqslant 51$; $0\leqslant J\leqslant 9$, are plotted in Fig. 3, in which the upper dashed line represents the energy of the $D\,^1\Sigma^+$ dissociation limit, H($1s$)+Li($3p$), located at 51 213.08 cm^{-1} above the potential minimum of the $X\,^1\Sigma^+$ state. Two term values, $T(51,10)$ and $T(51,11)$, are also outside the scale of the figure.

For each vibrational level with more than two rotational levels observed, effective rotational and vibrational constants, $B_{v,O}$ and $G_{v,O}$, are derived from a least-squared fit and listed in Table I, in which also included are (a) the $B_{v,O}$ and $G_{v,O}$ values for the $v\leqslant 25$ vibrational levels,[19] and (b) vibronic energies $G_{v,V}$, observed spectral linewidths, $\Gamma_{v,O}$, and vibronic resonance widths, $\Gamma_{v,V}$, to be further discussed in Sec. III C 2. For the $v\leqslant 25$ vibrational levels, several $B_{v,O}$ and $G_{v,O}$ values were slightly corrected after more transitions were observed. Particularly, (i) the $B_{3,O}$ value was corrected as 0.212 4(70) cm^{-1} because the term values fitted were of $J=1-7$, not just of $J=5-7$;[19] (ii) the $B_{4,O}$ value was corrected as 0.270 6(634) cm^{-1} because only the $J=4-6$ term values were fitted due to the lack of reproducibility of the $J=3$ level; (iii) the $B_{7,O}$ value was corrected as 1.800 0(1126) cm^{-1}, because only the $J=1-3$ term values were fitted due to the reassignment of the $J=4$ level to the level $v=8$, which is comparable to the adiabatic value of 1.84 cm^{-1} reported previously.[19] It is also noted that, (i) for the $v=4$ and 9 levels, we could only observe the $J\geqslant 4$ rotational levels; (ii) for the $v=6$ level, the only term value of 47 971.07 cm^{-1} (see Table III of Ref. 19) is now assigned as $D(6,7)$ after it was also observed from $B(0,7,f)$; and (iii) for the $v=7$ level, we found only the $J=1-3$ rotational levels. The vibrational level previously assigned as $v=32$ was reassigned as $v=31$. We failed to identify any level beyond the level $v=51$ of the $D\,^1\Sigma^+$ state, although the spectral range was scanned over the dissociation limit.

Note that we observed a few new vibrational levels having larger effective rotational constants and they were assigned to the levels among the four electronic states $I\,^1\Pi$, $E\,^1\Sigma^+$, $J\,^1\Pi$, and $K\,^1\Delta$,[27] which were predicted to be located near the inner, upper part of the $D\,^1\Sigma^+$ state. Among observed rovibrational levels, the lowest one is assigned as the $E\,^1\Sigma^+(0,0)$ level and located at 50 273.71 cm^{-1}, lying between the $v=41$ and 42 vibrational levels of the $D\,^1\Sigma^+$ state. The spectral results will be presented in a forthcoming paper.

C. The vibronic approach versus the adiabatic approach

The vibrational energies for the adiabatic levels have been calculated with the adiabatic *ab initio* $D\,^1\Sigma^+$ potential. The vibronic energies for the vibronic levels and their vibronic resonance widths have been obtained by a diabatic approach, in which the calculations of potential couplings involve a set of eight diabatic *ab initio* potentials of $^1\Sigma^+$ symmetry, corresponding to seven neutral asymptotes [Li($2s,2p,3s,3p,3d,4s,4p$)H($1s$)] and one ionic asymptote (Li^+H^-). Note that the details of the calculations will not be repeated here; the interested readers should refer to Refs. 29 and 30. For brevity, the comparisons of the vibrational energy spacings, the spectral linewidths, and the asymmetric line shapes are separately discussed below.

1. Vibrational energy spacing

Observed vibrational energy spacings ($\Delta G_{v+1/2}=G_{v+1}-G_v$, denoted as $\Delta G_{v+1/2,O}$), vibronic vibrational energy spacings ($\Delta G_{v+1/2,V}$), and adiabatic vibrational energy spacings ($\Delta G_{v+1/2,A}$) are derived and presented together in Table II. Note that the vibronic $\Delta G_{v+1/2,V}$ value at $v=3$ is corrected as 58.95 cm^{-1}. The lowest levels in the inner and outer wells of the $D\,^1\Sigma^+$ state correspond to quite different vibrational spacings and rotational constants ($\sim$250 and $\sim$1 cm^{-1} for the inner well and $\sim$59 and $\sim$0.2 cm^{-1} for the outer well). Since only one inner well level ($v=7$) is observed, our best estimate for the lowest inner well spacing is our vibronic calculation of the energy difference between levels $v=1$ and $v=7$. For the lowest outer well spacing, the energy level difference between $v=0$ and $v=2$ is observed to be 59.12 cm^{-1} and calculated vibronically to be 58.72 cm^{-1}. Note that $\Delta G_{6+1/2,V}$ is taken as the vibronic energy level difference between $v=6$ and $v=8$, and $\Delta G_{6+1/2,A}$ the adiabatic energy level difference between $v=6$ and $v=8$. Theoretical calculations predict that there are 59 adiabatic vibrational levels, and 55 effective vibrational levels after an extensive vibronic treatment.

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TABLE I. Observed rotational constants, $B_{v,O}$ (in cm^{-1}), observed and vibronic vibrational energies $G_{v,O}$ and $G_{v,V}$ (in cm^{-1}) for vibrational levels of the ^{7}LiH $D\,^1\Sigma^+$ state, and the observed and vibronic spectral linewidths (HWHM), $\Gamma_{v,O}$ and $\Gamma_{v,V}$ (in cm^{-1}), $\Gamma_{v,O}$ at $J=5$ and $\Gamma_{v,V}$ at $J=0$. The parenthesized number represents one standard deviation.

v	$B_{v,O}$	$G_{v,O}$	$G_{v,V}$[a]	$\Gamma_{v,O}$	$\Gamma_{v,V}$	Reference[b]
			(Primarily inner well)			
1	⋯	⋯	477 75.85	⋯	12.021	β
7	1.800 0(1126)	479 88.88(88)	480 03.18	0.60	2.721	α, β
			(Primarily outer well)			
0	0.196 5(15)	476 67.45(3)	476 76.77	0.10	0.232	α, β
2	0.210 6(13)	477 26.57(3)	477 35.49	0.15	0.0085	α, β
3	0.212 4(70)	477 86.05(15)	477 93.61	0.16	0.323	α, β
4	0.270 6(634)	478 43.44(203)	478 52.56	0.22	0.377	α, β
5	0.309 8(25)	479 00.40(2)	479 07.76	0.20	1.207	α, β
6	[c]	[c]	479 63.82	⋯	0.586	β
8	0.311 1(37)	480 07.26(4)	480 12.99	0.10	0.793	α, β
9	0.634 8(153)	480 47.73(47)	480 60.24	0.15	0.997	α, β
			(Above the intermediate barrier)			
10	⋯	⋯	480 96.35	⋯	0.789	β
11	0.748 6(6)	481 16.77(1)	481 24.98	0.50	2.604	α, β
12	0.780 7(22)	481 62.20(2)	481 68.57	0.15	0.588	α, β
13	0.859 6(763)	482 03.32(193)	482 16.54	0.30	0.766	α, β
14	0.747 4(790)	482 48.10(184)	482 59.37	1.25	6.793	α, β
15	0.791 0(52)	483 06.18(8)	483 10.57	0.11	0.00004	α, β
16	0.827 4(36)	483 64.24(3)	483 65.86	0.30	0.412	α, β
17	⋯	⋯	484 28.13	⋯	4.893	β
18	0.888 1(697)	484 57.87(162)	484 71.05	1.25	12.595	α, β
19	0.821 2(13)	485 31.48(1)	485 32.00	0.11	0.290	α, β
20	0.836 3(14)	485 98.60(1)	485 98.08	0.12	0.259	α, β
21	1.102 5(306)	486 75.12(42)	486 68.66	3.0	1.02	β, γ
22	⋯	⋯	487 39.70	⋯	37.582	β
23	0.849 1(34)	487 69.15(24)	487 62.58	3.5	9.918	α, β
24	0.834 2(63)	488 49.49(7)	488 45.45	0.30	0.755	α, β
25	0.825 2(19)	489 26.82(2)	489 21.99	0.12	0.023	α, β
26	0.848 5(15)	490 06.83(3)	490 00.19	0.50	0.313	γ
27	0.823 2(97)	490 91.45(14)	490 83.26	1.0	1.543	γ
28	0.831 7(55)	491 76.97(8)	491 68.83	1.0	1.853	γ
29	0.788 0(65)	492 62.09(9)	492 53.27	1.0	1.384	γ
30	0.795 8(46)	493 46.49(5)	493 37.26	0.30	1.094	γ
31	0.787 5(48)	494 30.52(6)	494 21.42	0.45	0.655	γ
32	0.779 3(11)	495 12.12(1)	495 04.52	0.20	0.145	γ
33	0.770 0(15)	495 92.07(1)	495 85.62	0.15	0.00085	γ
34	0.765 4(18)	496 71.79(2)	496 65.28	0.10	0.0011	γ
35	0.748 9(54)	497 53.42(8)	497 45.10	0.50	0.154	γ
36	0.893 4(216)	498 32.72(50)	498 26.60	2.0	1.290	γ
37	0.693 0(216)	499 25.13(45)	499 11.20	1.5	3.460	γ
38	0.740 0(120)	500 10.07(14)	499 98.18	0.75	3.171	γ
39	0.727 6(49)	500 91.73(7)	500 81.78	0.15	0.829	γ
40	0.694 1(132)	501 69.95(25)	501 61.60	0.50	0.0049	γ
41	0.637 0(112)	502 44.94(25)	502 38.40	1.0	0.829	γ
42	0.670 0(38)	503 16.73(8)	503 12.42	1.5	1.858	γ
43	0.685 9(145)	503 90.07(23)	503 84.96	0.50	1.668	γ
44	0.663 7(13)	504 65.48(2)	504 58.14	0.20	0.522	γ
45	0.670 7(68)	505 41.50(10)	505 32.58	1.5	0.013	γ
46	0.703 9(174)	506 17.75(37)	506 07.91	3.0	1.157	γ
47	0.538 0(887)	506 99.81(200)	506 59.55	6.0[d]	0.037	γ
48	0.598 4(347)	507 88.60(59)	507 66.04	7.5[d]	8.947	γ
49	0.450 8(133)	508 60.63(21)	508 48.16	1.5	5.900	γ
50	0.604 0(33)	509 32.87(5)	509 23.49	0.25	2.022	γ
51	0.599 3(17)	510 04.32(3)	509 95.80	0.35	0.210	γ
52	⋯	⋯	510 66.10	⋯	0.169	γ
53	⋯	⋯	511 34.18	⋯	1.520	γ
54	⋯	⋯	511 97.02	⋯	3.294	γ

[a]With respect to the potential minimum of the ^{7}LiH $X\,^1\Sigma^+$ state.
[b](α) see also Ref. 19; (β) see also Ref. 30; (γ) This work.
[c]A single level $D(6,7)$ located at 47 971.07 cm^{-1}.
[d]Estimated values, see the text for details.

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TABLE II. The vibrational energy spacings of the $D\,^1\Sigma^+$ state. The v value represents the vibrational quantum number. $\Delta G_{v+1/2,O}$, $\Delta G_{v+1/2,V}$ and $\Delta G_{v1/2,A}$ are the observed, vibronic, and adiabatic vibrational energy spacings, respectively.

v	$\Delta G_{v+1/2,O}$	$\Delta G_{v+1/2,V}$	$\Delta G_{v+1/2,A}$
	(Primarily inner well)		
1	⋯	227.33	272.44
7	⋯	⋯	⋯
	(Primarily outer well and above the intermediate barrier)		
0	59.12	58.72	57.46
2	59.48	58.12	57.81
3	57.39	58.95	57.28
4	56.96	55.21	56.20
5	⋯	56.06	54.34
6	⋯	49.17	51.39
8	40.47	47.25	45.33
9	⋯	36.11	28.77
10	⋯	28.63	30.05
11	45.43	43.60	40.70
12	41.12	47.97	43.84
13	44.78	42.83	46.78
14	58.08	51.20	49.43
15	58.06	55.29	51.76
16	⋯	62.27	53.85
17	⋯	42.92	55.73
18	73.61	60.94	57.48
19	67.12	66.09	59.14
20	76.52	70.57	60.73
21	⋯	71.04	62.21
22	⋯	22.88	63.55
23	79.92	82.87	64.80
24	77.33	76.54	65.89
25	80.01	78.20	66.88
26	84.62	83.07	67.77
27	85.52	85.57	68.54
28	85.12	84.44	69.24
29	84.40	83.99	69.86
30	84.03	84.16	70.40
31	81.60	83.10	70.85
32	79.95	81.10	71.24
33	79.72	79.66	71.54
34	81.63	79.82	71.77
35	79.30	81.50	71.92
36	92.41	84.60	72.02
37	84.94	86.98	72.04
38	81.66	83.60	72.01
39	78.22	79.82	71.94
40	74.99	76.80	71.82
41	71.79	74.02	71.65
42	73.34	72.54	71.45
43	75.41	73.18	71.21
44	76.02	74.44	70.93
45	76.25	75.33	70.61
46	82.06	51.64	70.26
47	88.79	106.49	69.88
48	72.03	82.12	69.47
49	72.24	75.33	69.02
50	71.45	72.31	68.55
51		70.30	68.05
52		68.08	67.51
53		62.84	66.91
54			66.25
55			65.39
56			64.02
57			60.73

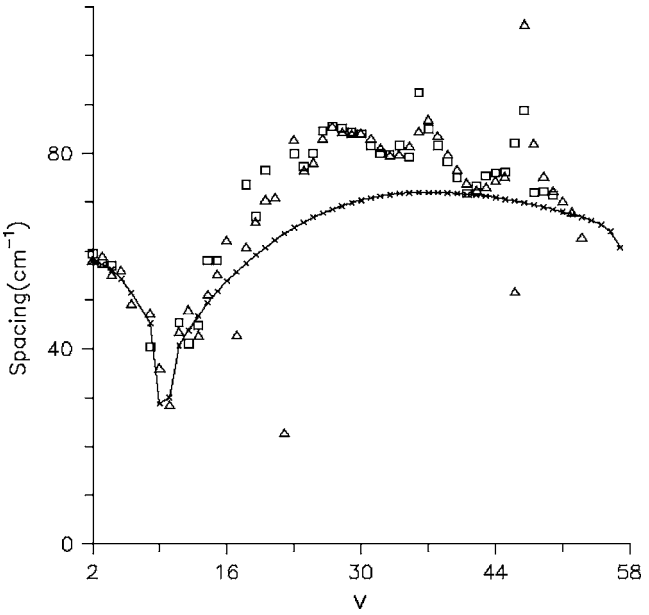


FIG. 4. Plot of vibrational energy spacings $\Delta G_{v+1/2,O}$ (open squares), $\Delta G_{v+1/2,V}$ (open triangles), and $\Delta G_{v+1/2,A}$ (crosses) with respect to the vibrational quantum number v.

The variation of the vibrational energy spacing with respect to the vibrational quantum number is shown in Fig. 4, in which only the $\Delta G_{v+1/2}$ values for $v \geqslant 2$ are portrayed, and the $\Delta G_{v+1/2,A}$ values (by the symbol crosses) are connected with a solid line for the purpose of comparison. One can tell from Fig. 4 that, for the vibrational levels with $v > 10$, the variation of adiabatic vibrational energy spacing ($\Delta G_{v+1/2,A}$) is relatively small and regular, while the variation of vibronic vibrational energy spacing ($\Delta G_{v+1/2,V}$, open triangles) is relatively large and irregular. It is found that the variation of the observed vibrational energy spacing ($\Delta G_{v+1/2,O}$, open squares) is irregular and similar to that for the vibronic vibrational energy spacing although their relative values are somewhat different. Such a similar irregularity demonstrates that the D state is indeed best characterized by the vibronic approach.

It should also be noted that, in both Table II and Fig. 4, the same character v was used to identify each vibrational level; however, its physical meaning would depend upon whether the potential is adiabatic or not. In an adiabatic potential, it represents the traditional vibrational number, but it should be taken as an effective one after the vibronic treatment.

2. Spectral linewidth

Previous work[19,30] demonstrated that (i) observed spectral linewidths and theoretical resonance widths vary with the effective vibrational quantum number among the lowest 26 ($v \leqslant 25$) vibrational levels; and (ii) the vibronic approach is the favored approach for the D state.

One would expect that OODR spectral linewidths observed for the higher ($v > 25$) vibrational levels also vary if the vibronic approach is indeed favored for the D state. In Fig. 2, the spectrum (a) is composed of two broad OODR-type transitions, D-$A(38,10)$ $P3$ and $R3$, and one narrow

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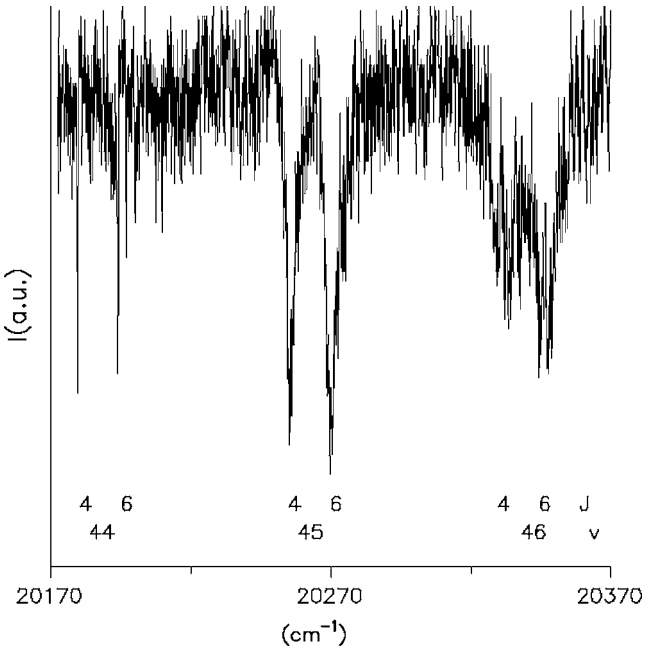


FIG. 5. An OODR spectrum of ^{7}LiH. The six spectral dips represent the D-$A(44-46,10)$ $P5$ and $R5$ OODR transitions.

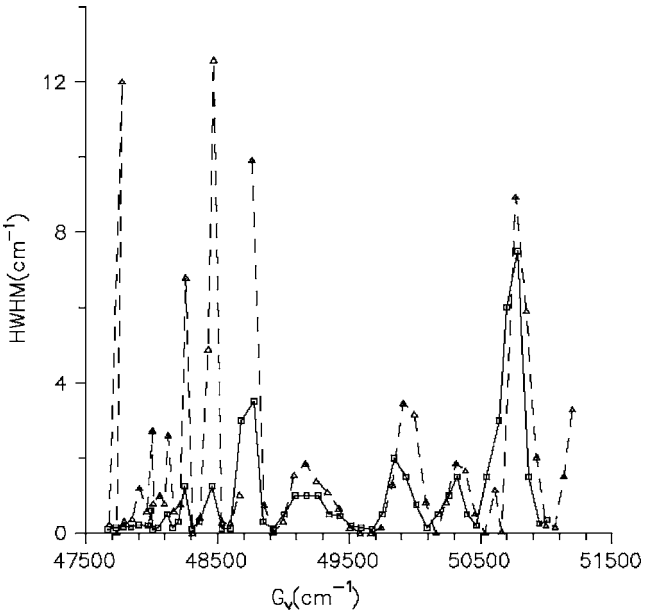


FIG. 6. Plot of observed spectral linewidths (open squares) and calculated nonradiative resonance widths (open triangles) (HWHM in cm^{-1}) with respect to the vibrational energy G_v.

SEP-type transition, A-$X(10,8)$ $P4$; and the spectrum (b) two narrow OODR-type transitions, D-$A(39,10)$ $P3$ and $R3$, and one narrow SEP-type transition, A-$X(10,8)$ $R2$. In Fig. 5, the observed spectrum is composed of six OODR-type transitions: two narrow ones, D-$A(44,10)$ $P5$ and $R5$; and four broad ones, D-$A(45,10)$ $P5$ and $R5$, and D-$A(46,10)$ $P5$ and $R5$. These three spectra demonstrate that the spectral linewidths among newly observed vibrational levels do vary with the effective vibrational quantum number.

Forty-seven overall observed (not corrected for the Doppler effect) spectral linewidths (HWHM, $\Gamma_{v,O}$) and 55 vibronic resonance widths ($\Gamma_{v,V}$) were also listed in Table I, in which each $\Gamma_{v,O}$ value was given with two significant digits. The vibronic resonance widths are those calculated at $J=0$, but observed spectral linewidths are mostly those measured at $J=5$, the rotational level identified for most observed vibrational levels. It should be noted that for the vibrational levels $v=47$ and 48, $\Gamma_{v,O}$ was estimated by taking one-half the measured spectral linewidth (HWHM) since the P and R transitions for each relevant J were found overlapped as a single broad peak. For visualizing the variation, observed spectral linewidths (open squares and solid line) and vibronic resonance widths (open triangles and dashed line) are portrayed in Fig. 6, in which the $\Gamma_{v,V}$ value at $v=22$ is 37.582 cm^{-1}, which is outside the scale of the figure. One can tell from Fig. 6 that for the vibrational levels with $v>10$, the observed spectral linewidths and the vibronic resonance widths vary similarly with the vibrational energy G_v. It is also noted that (i) the two long-lived vibronic states, predicted at $v=40$ and 45, are each just one vibrational level higher than the two narrow-linewidth levels, observed at $v=39$ and 44; and (ii) the likely long-lived vibronic state predicted at $v=47$ does not correspond to any observed level, in terms of either linewidth or energy shown in Table I. In sum, both observed spectral linewidths and vibronic resonance widths follow a similar variation with respect to the effective vibrational quantum number. Such a similar variation demonstrates again that the D state is indeed better characterized by the vibronic approach.

Note that among the $v\leqslant 25$ vibrational levels, the levels $v=1$ and 22 might not be observable since their vibronic resonance widths are larger than the highest linewidth of 7.5 cm^{-1} observed for the level $v=48$. Both (a) a single $D(6,7)$ level observation and (b) the failure of observing the levels $v=10$ and 17 seem to indicate that the one-to-one correspondence[30] between the adiabatic and the vibronic levels for the case of $v\leqslant 25$ is doubtful, as compared to the fact that, among the $v>25$ adiabatic vibrational levels, four do not exist any more after the vibronic treatment.

3. Spectral line shape

During the OODR experiments, we found that some spectral peaks have an asymmetric lineshape, a Fano-type profile.[36–39] Two of observed OODR spectra are reproduced in Figure 7, in which the spectrum (a) corresponds to the D-$A(50,8)$ $P6$ transition, and the spectrum (b) the D-$A(51,9)$ $P6$ transition.

Since a quantitative simulation is out of the scope of the present work, only a qualitative interpretation is described here. Gadea and Boutalib[40] calculated the radial couplings between the $^{1,3}\Sigma^{+}$ states of this molecule. The D-C radial coupling was predicted to be very strong. Thus the continuum states of the C state could couple with some bound levels of the D state to form corresponding modified states. For each OODR spectral peak, the Fano profile index (q), and in turn the spectral line shape, will be determined by the relative magnitudes of (i) the coupling strength between the D discrete level and the C continuum states, (ii) the transition dipole moment between the modified state and the A discrete level, and (iii) the transition dipole moment between

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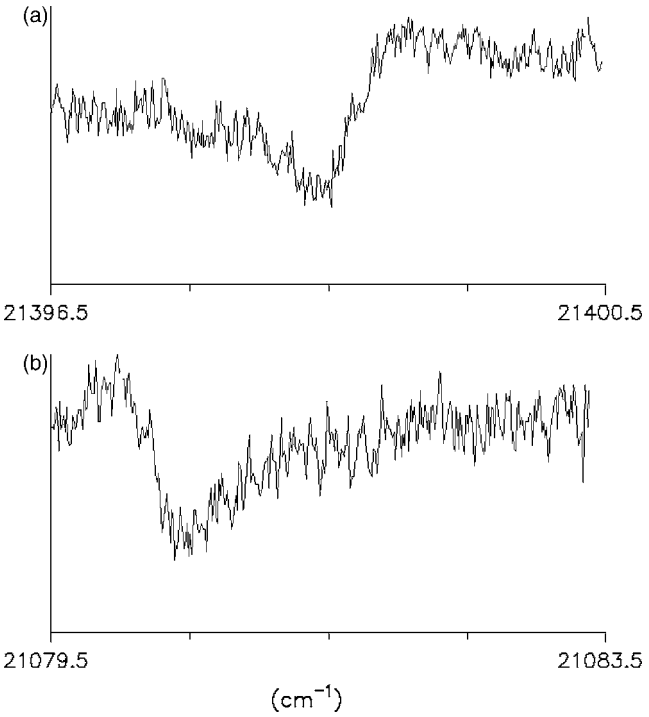


FIG. 7. OODR spectra of ^{7}LiH. The spectrum (a) corresponds to the D-$A(50,8)$ $P6$ transition, and the spectrum (b) the D-$A(51,9)$ $P6$ transition.

the C continuum states and the A state discrete level. In sum, the observation of Fano-type spectral lines demonstrates that a strong radial coupling occurs between the C and D states and, in turn, it contributes partially to the fact that the D state is characterized by the vibronic approach.

IV. CONCLUSIONS

The observation of the excited electronic state $D\,^1\Sigma^+$ of ^{7}LiH has been extended close to its dissociation limit by a pulsed optical-optical double resonance fluorescence depletion spectroscopic technique. An extensive vibronic calculation has also been performed with consistent diabatic and adiabatic *ab initio* potentials. The 292 new OODR transitions, corresponding to 197 rovibrational levels among 26 vibrational levels, have been observed. Totally, 47 vibrational states have been observed for the D state. Observed vibrational energy spacings and the vibronic vibrational energy spacings, not the adiabatic ones, are similarly irregular. Observed spectral linewidths and the vibronic resonance widths are found to follow a similar variation with respect to the effective vibrational quantum number. The observation of asymmetric spectral line shape may be attributed to the radial couplings between the discrete levels of the D state and the continuum states of the C state. In sum, (i) the result thatboth observed and vibronic vibrational energy spacings are similarly irregular, (ii) the result that both spectral and vibronic linewidths follow a similar variation, and (iii) the observation of asymmetric line shape confirm altogether that the D state is better characterized by the vibronic approach.

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